

## Real-Time Traffic Density Management System

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### Abstract

Traffic congestion is a critical issue in urban areas, leading to increased travel time, fuel consumption, and environmental pollution. Traditional traffic light systems operate on fixed time cycles, which often fail to adapt to real-time traffic conditions, causing unnecessary delays. This paper presents a real-time traffic density management system that dynamically adjusts traffic light durations based on vehicle density and priority. The proposed system integrates ultrasonic sensors (HC-SR04), RFID technology (RC522 module), and Arduino microcontrollers (Mega 2560 & Nano) to optimize traffic flow efficiently. The system consists of four ultrasonic sensors placed at different lanes to measure vehicle density in real-time. Based on the data collected, an Arduino-based algorithm determines the optimal signal timing for each lane, reducing congestion. Additionally, RFID technology is implemented to detect emergency vehicles (such as ambulances, fire trucks, and police vehicles), ensuring immediate signal clearance for high-priority vehicles. The signal control is achieved using LED indicators and a signal light module, powered by a 12V DC power supply. This research aims to enhance urban traffic management by offering an automated, cost-effective, and scalable solution. The experimental results demonstrate that the proposed system significantly improves traffic flow efficiency, reduces waiting time, and prioritizes emergency vehicles effectively. Future enhancements could include integrating machine learning algorithms and IoT connectivity for advanced predictive traffic control.

**Keywords-** Arduino mega 2560, Arduino nano, RFID technology, Smart traffic system, Traffic management, Ultrasonic sensors, Vehicle priority

### INTRODUCTION

Traffic congestion is a persistent problem in modern cities, leading to delays, higher fuel consumption, and increased air pollution. Conventional traffic light system operates on fixed timings, irrespective of real-time traffic conditions, often resulting in unnecessary delays. The density-priority traffic system aims to address these inefficiencies by implementing a smart traffic management solution that prioritizes lanes based on vehicle density. The system uses sensors to gather real-time data,

processes this information using a microcontroller, and dynamically adjusts signal timings to ensure smoother traffic flow.

Each functionality, from density based traffic control to emergency vehicle prioritization can be developed incrementally and tested independently. Prototyping is simple with small scales intersection models, and real-time feedback simplifies debugging and optimization. Its practical relevance and adaptability make it a user friendly project for addressing real world traffic challenges effectively. These systems are particularly

beneficial in urban areas where traffic density and complexity are high. One of the key features of a smart traffic light system is its ability to detect vehicle density at intersections.

Smart traffic lights also incorporate priority mechanisms to facilitate the movement of emergency vehicles such as ambulances, fire trucks, police cars. Radio-frequency identification (RFID) based tracking systems can be integrated to identify these vehicles in real-time and alter traffic signals to provide them with a clear path.

### LITERATURE SURVEY

Santanu Agarwal et al. discuss the inefficiencies of traditional traffic management systems and the delays faced by emergency vehicles, which motivates the use of IoT to improve traffic flow. Their proposed system uses IoT to integrate density-based traffic signal automation and manual override via the internet, ensuring timely passage for emergency vehicles and minimizing congestion-related delays [1].

Dayanidhi Krishna et al. introduce a traffic management system that dynamically adjusts signal timings based on traffic density. The system employs a pixel-density-based method to evaluate road congestion, making it cost-effective and suitable for developing nations [2]. The algorithm reallocates time from less congested lanes to denser ones, maintaining cost-efficiency with Arduino integration [2].

Rahul N. Dhole et al. propose a smart traffic signal that uses ultrasonic sensors to monitor and manage vehicle flow. The system dynamically adjusts signal timings based on real-time data collected through these sensors, aiming to reduce congestion. This offers a cost-efficient solution compared to traditional static signal systems [3].

### METHODOLOGY

The methodology for a smart traffic light system involves designing,

implementing, and testing a system that dynamically adjusts traffic signals based on vehicle density and prioritizes emergency vehicles. The process is divided into several stages, each involving the integration of hardware and system design and planning.

The project begins with the design of the system architecture. The traffic light system will consist of multiple components, including an Arduino mega2560, Arduino nano, ultrasonic sensors, RFID modules, LEDs, resistors, and a signal light module PCB. The road intersection is simulated on a solderless breadboard, with four lanes equipped with traffic lights and sensors. The system layout ensures that sensors are positioned to detect vehicle density on each lane while RFID modules are placed strategically to detect emergency vehicles equipped with RFID tags [4].

#### Hardware Integration

##### *Ultrasonic Sensors (HC-SR04)*

In Fig. 1, ultrasonic sensors (HC-SR04) are placed on each lane to measure the distance of vehicles from the intersection. By analysing the distance data, the system determines the density of vehicles in each lane. The lane with the highest density is given priority by extending the green light duration [3].



**Figure 1:** HC-SR04.

##### *RFID Modules (RC522)*

In Fig. 2, RFID modules and tags are used to detect and prioritize emergency

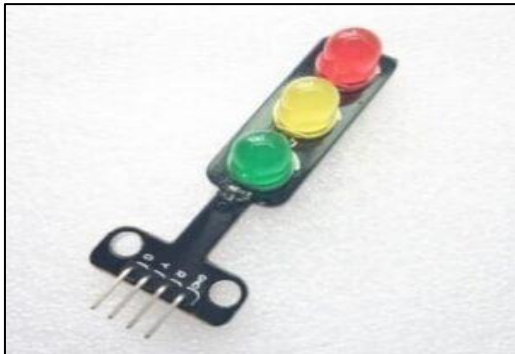
vehicles. When an RFID tag is detected, the system immediately switches the signal to green for the corresponding lane, ensuring a clear path.



**Figure 2:** RFID modules (RC522).

### **Traffic Light Components**

In Fig. 3, red, orange, and green LEDs, along with resistors, simulate the traffic lights. These components are controlled by the Arduino boards to manage the signal timings based on the sensor data [5], [6].

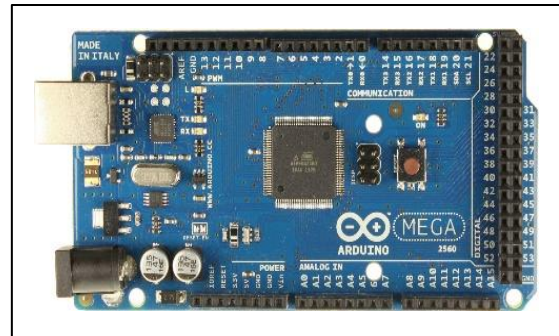


**Figure 3:** Signal light module PCB.

### **Microcontrollers**

In Fig. 4, red, orange, and green LEDs, along with resistors, simulate the traffic lights. These components are controlled by the Arduino boards to manage the signal timings based on the sensor data [3], [6].

The Arduino Mega 2560 is used as the primary controller for managing sensor data, prioritizing lanes, and controlling traffic lights. The Arduino Nano handles specific tasks like processing RFID data.



**Figure 4:** Arduino Mega 2560.

### **Software Development**

The system is programmed using the Arduino IDE. The code includes:

- **Signal Timing Algorithm:** A dynamic algorithm adjusts signal timings based on vehicle density. If two or more lanes have similar densities, the system cyclically rotates the green light.
- **Priority Management:** The RFID module detects emergency vehicle tags, and the system overrides normal signal operations to provide a green light to the emergency vehicle's lane.
- **Synchronization:** The system ensures proper synchronization between all lanes, avoiding conflicts and ensuring smooth traffic flow.

### **Power Supply Setup**

A 12V DC power supply is used to power the microcontrollers and other components. Proper connections are made using jumper wires to ensure reliable and safe operation.

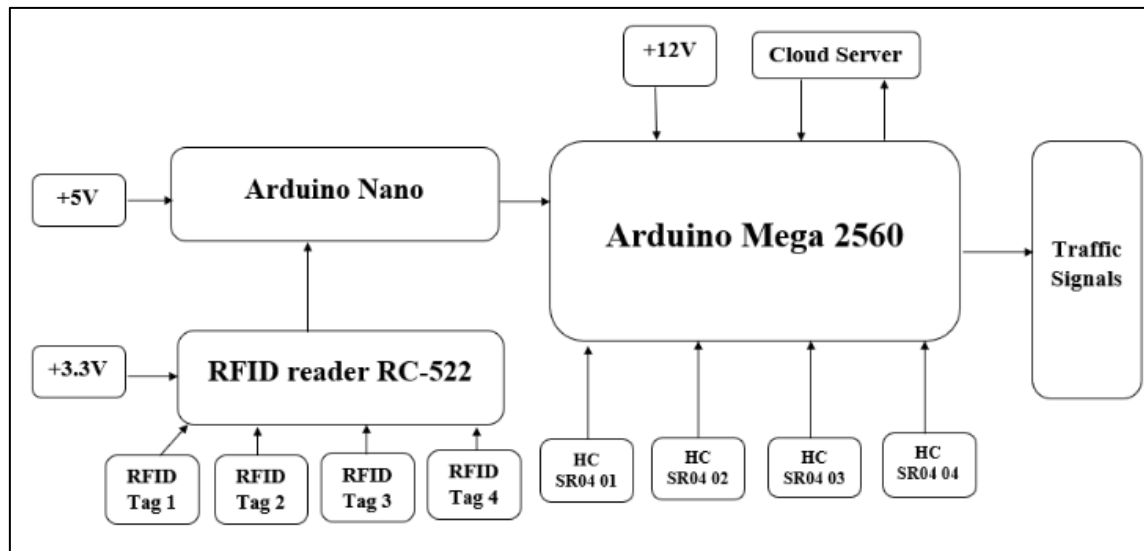
### **System Testing and Optimization**

The prototype is tested in a simulated environment to verify its functionality. Different scenarios such as varying vehicle densities and emergency vehicle detection, are simulated to observe system's performance. The system is optimized by fine-tuning the signal timing algorithm and ensuring seamless interaction between sensors, RFID modules and traffic lights.

## Implementation and Results

The final stage involves implementing the complete system and evaluating its effectiveness in reducing

congestion and prioritizing emergency vehicles. The results are documented to assess the system's efficiency, response time, and reliability.



**Figure 5:** Block diagram of smart traffic light system.

The block diagram in Fig. 5 represents a smart traffic light control system that integrates RFID-based priority detection and ultrasonic sensor-based traffic density analysis. The system is built using Arduino Mega 2560 and Arduino Nano as the main controllers.

### Power Supply Section

+12V is supplied to the Arduino Mega 2560, which serves as the microcontroller. +5V powers the Arduino Nano, responsible for handling RFID-based vehicle priority detection. 3.3V is supplied to the RFID Reader (RC-522) for scanning RFID tags.

### Arduino Mega 2560 (Main Controller)

The Arduino Mega 2560 is responsible for controlling the traffic lights based on inputs from ultrasonic sensors and the Arduino Nano (RC-522) for scanning and Arduino Nano (RFID priority system). It receives traffic density data from the four

HC-Sr04 ultrasonic sensors, placed at each lane to detect vehicle congestion. Based on the sensor data, the traffic signal timings are dynamically adjusted to reduce waiting time.

### Arduino Nano (RFID Priority System)

The Arduino Nano is dedicated to RFID-based vehicle priority detection. It processes RFID data from the RC-522 RFID Reader and checks the authenticity of passing vehicles using four RFID tags assigned to priority vehicles such as Ambulance, police cars, or VIP vehicles. Once a priority vehicle is detected, the Arduino Nano sends a signal to the Arduino Mega 2560 to override normal traffic rules and give immediate clearance [7].

### RFID Reader (RC-522) and RFID Tags

The RFID Reader (RC-522) operates at 3.3V and communicates with the Arduino Nano to scan passing the RFID tags. Four RFID tags are assigned to different priority vehicles, allowing automatic detection when

they approach the intersection [7].

### HC-SR04 Ultrasonic Sensors (Traffic Density Detection)

Four HC-SR04 sensors are placed on different lanes to detect traffic density based on the distance between vehicles. The sensor data is processed by Arduino Mega 2560, which adjusts the traffic light timing, dynamically allocating more green light time

to congested lanes.

### Traffic Light Control System

The Arduino Mega 2560 controls four sets of traffic lights (each with red, yellow, and green LEDs). The lights change based on traffic density analysis and priority vehicle detection. Table 1 illustrates the comparison between smart traffic system and normal traffic system.

**Table 1:** Comparison between smart traffic system and normal traffic system.

| Smart Traffic System                                                      | Normal Traffic System                                      |
|---------------------------------------------------------------------------|------------------------------------------------------------|
| Can detect emergency vehicles and change lights accordingly.              | Emergency vehicles may have to wait at red lights.         |
| More energy efficient due to adaptive control.                            | May waste energy by running lights unnecessarily.          |
| Can integrate by optimizing signal changes.                               | Limited to basic traffic control.                          |
| Reduces carbon emissions by optimizing traffic flow.                      | Increase emissions due to stop-and-go traffic patterns.    |
| Collects and analyses real-time traffic data for continuous optimization. | No data collection or learning from past traffic patterns. |

Table 1 illustrates the key differences between smart and normal traffic systems. As shown in Fig. 6, smart traffic systems offer significant advantages over traditional systems, including improved emergency vehicle handling, energy efficiency, and reduced carbon emissions.

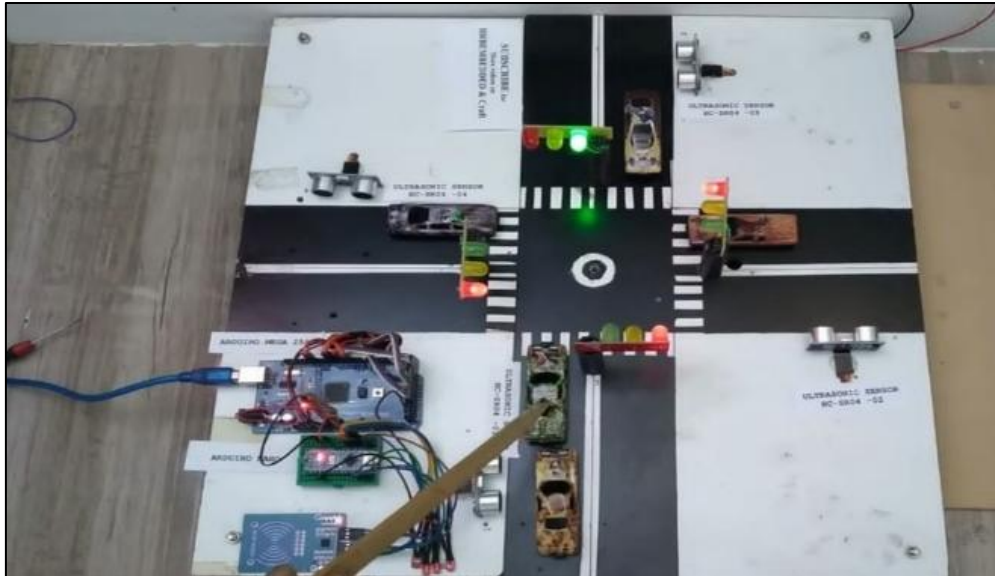
## RESULT

The proposed smart traffic light system was successfully implemented and tested under various traffic conditions. The system dynamically adjusted traffic signal durations based on real-time vehicle density and provided priority clearance for emergency vehicles. During testing, the system effectively detected traffic congestion and adjusted the green light duration accordingly. Compared to traditional fixed-time traffic signals, the system demonstrated a significant reduction in waiting time and congestion, particularly during peak hours. The priority mechanism for emergency vehicles was also tested,

ensuring immediate clearance when a high-priority vehicle was detected. This feature significantly reduced delays for ambulances and other essential services, enhancing response times and road safety. The system functioned reliably throughout the testing phase, with smooth transitions between signal states and real-time adjustments to changing traffic conditions. The results indicate that the proposed model can effectively reduce traffic congestion, optimize signal timing, and improve emergency vehicle mobility.

Future enhancements could include machine learning-based predictive traffic control to further optimize traffic flow and IoT integration for real-time data sharing with traffic management systems. These improvements would enhance the system's adaptability and scalability for smart city applications. The results confirm that the Smart Traffic Light System is a viable and effective solution for modern traffic management, contributing to safer and more efficient urban transportation.





**Figure 6:** Comparison of smart and normal traffic systems.

## CONCLUSION

The smart traffic light system we have designed aims to enhance urban traffic management by incorporating vehicle density monitoring and priority-based control. Traditional traffic signals operate on fixed-time cycles, often leading to unnecessary delays and congestion. Our system overcomes these limitations by dynamically adjusting signal timings based on real-time traffic conditions, improving overall efficiency.

Using Arduino Mega 2560 and Arduino Nano, along with HC-SR04 ultrasonic sensors, our system detects the number of vehicles at an intersection. This data is processed to determine optimal green light durations, ensuring a smooth traffic flow. Additionally, an RFID-based priority mechanism is integrated to give emergency and high-priority vehicles, such as ambulances and fire trucks, immediate passage. The RFID RC522 module and RFID tags facilitate automatic vehicle identification, allowing the system to override standard signals when necessary. The implementation of LED indicators, resistors, and a signal light module ensures effective communication of traffic status to drivers. A 12V DC power supply is used to support the hardware components, making

the system energy-efficient and practical for real-world deployment.

The proposed system provides multiple benefits, including reduced congestion, decreased fuel consumption, and improved emergency response times. Unlike traditional traffic lights, which rely on predetermined cycles, our intelligent traffic system adapts dynamically, leading to better traffic flow and enhanced road safety. Future enhancements could include the integration of machine learning algorithms for predictive traffic analysis and IoT-based connectivity for real-time data sharing with central traffic management systems. Additionally, expanding the system to accommodate pedestrian crossings and smart city networks could further improve urban mobility. In conclusion, our smart traffic light system represents a significant step toward intelligent, responsive, and efficient traffic management, contributing to the development of smarter and safer cities.

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